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# Step 1: Import required libraries
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Step 2: Upload an image
uploaded = files.upload()

# Step 3: Read the uploaded image
filename = list(uploaded.keys())[0]
image = cv2.imread(filename)

# Step 4: Define Laplacian mask with positive center coefficient
laplacian_mask = np.array([
    [-1, -1, -1],
    [-1,  8, -1],
    [-1, -1, -1]
], dtype=np.float32)

# Step 5: Apply the Laplacian mask
laplacian = cv2.filter2D(image, cv2.CV_32F, laplacian_mask)

# Step 6: Sharpen the image
sharpened = cv2.addWeighted(
    image.astype(np.float32), 1.0,
    laplacian, -1.0, 0
)

# Step 7: Convert the result to valid image range
sharpened = np.clip(sharpened, 0, 255).astype(np.uint8)

# Step 8: Convert BGR to RGB
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
sharpened_rgb = cv2.cvtColor(sharpened, cv2.COLOR_BGR2RGB)

# Step 9: Display original and sharpened images
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(sharpened_rgb)
plt.title("Sharpened Image - Positive Center Laplacian")
plt.axis("off")

plt.show()
```

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Original Image



Sharpened Image - Positive Center Laplacian

